

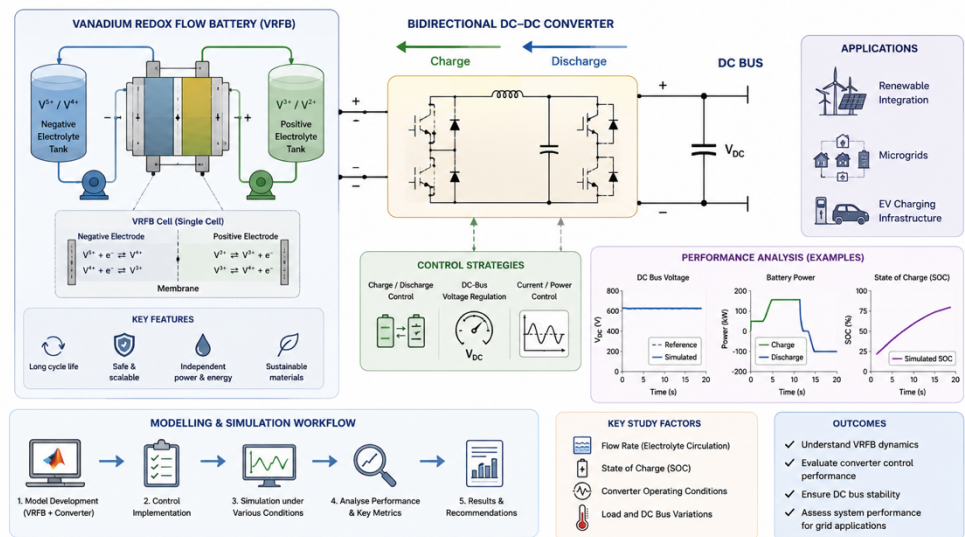
Student proposed?	Y/N	N
ID:	PB05	
SUPERVISOR:	BARENDSE	
TITLE:	Dynamic Modelling and Control of a Vanadium Redox Flow Battery Interfaced Through a Bidirectional DC-DC Converter	
DESCRIPTION:	Vanadium Redox Flow Batteries (VRFBs) are emerging as a promising technology for large-scale and long-duration energy storage, offering advantages in cycle life, safety and scalability over conventional battery technologies. Despite these benefits, their dynamic behaviour differs significantly from lithium-ion batteries due to the influence of electrolyte flow, electrochemical processes and auxiliary system requirements. This project investigates the dynamic modelling and control of a VRFB connected to a DC bus through a bidirectional DC-DC converter using MATLAB/Simulink. The student will utilise an existing VRFB model and develop the associated power electronic	

interface to study charging and discharging operation, DC-bus regulation and transient response.

The project aims to identify the key factors that influence VRFB performance and to assess their implications for future applications such as renewable energy integration, microgrids and electric vehicle charging infrastructure. Through simulation studies, the student will develop an understanding of the interaction between flow battery dynamics and converter control, providing a foundation for future research on advanced energy storage systems.

Dynamic Modelling and Control of a Vanadium Redox Flow Battery

Interfaced Through a Bidirectional DC-DC Converter



DELIVERABLES:

1. Literature review on vanadium redox flow batteries, battery energy storage systems and power electronic interfaces.
2. Development and validation of a MATLAB/Simulink model of a VRFB and bidirectional DC-DC converter system.
3. Implementation of charge/discharge control and DC-bus regulation strategies.
4. Investigation of system performance under varying load, state-of-charge and operating conditions.
5. Analysis of the influence of flow rate, state of charge and converter operating conditions on system performance.
6. Final report documenting the developed model, simulation results and recommendations for future grid-integration studies.

SKILLS/REQUIREMENTS:

MATLAB/Simulink modelling

- Power electronics and converter control
- Battery energy storage systems

GA 1: Problem solving:

Identify, formulate, analyse and solve complex* engineering problems creatively and innovatively

The project requires students to model and analyse the interaction between a VRFB and a bidirectional DC-DC converter, and to determine how battery dynamics influence power control, efficiency and DC-bus regulation.

GA 4*: Investigations, experiments and analysis: Demonstrate

The project involves simulation-based investigations of VRFB operation under varying operating conditions, including analysis of charge/discharge behaviour, transient response and converter performance.

competence to design and conduct investigations and experiments.	
GA 5: Use of engineering tools: <i>Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems</i>	The project requires the use of MATLAB/Simulink, battery modelling techniques, power electronic converter models and control-system analysis tools to evaluate the performance of an advanced energy storage system.
EXTRA INFORMATION:	The project will utilise an existing MATLAB/Simulink vanadium redox flow battery (VRFB) model available through the MathWorks Simscape Battery toolbox as a foundation for the study. The emphasis is on understanding the behaviour of flow battery technology and its interaction with power electronic converters rather than on detailed electrochemical model development. Students will extend the model through the addition of a bidirectional DC-DC converter and associated control strategies. Reference model: https://www.mathworks.com/help/simscape-battery/ug/analyze-performance-vanadium-redox-flow-battery.html [1] Bence Sziffer, Martin János Mayer, Viktor Józsa, "Detailed system modeling of a vanadium redox flow battery operating at various geographical locations", Applied Energy, Volume 384, 2025, 125473, ISSN 0306-2619. [2] X. Li, Y. Song, Y. Qiu, D. Mu and Y. Ning, "Modeling and Simulation of MWh Vanadium Redox Batteries System," 2023 IEEE 11th Joint International Information Technology and Artificial Intelligence Conference (ITAIC), Chongqing, China, 2023, pp. 1823-1830, doi: 10.1109/ITAIC58329.2023.10408756.
BROAD Research Area:	Power Engineering, Energy Storage Systems, Power Electronics
Project suitable for ME/ECE/EE/ALL?	ME/EE

Ethics clearance questionnaire

		Yes	No
Q1	Does this project involve data collection		X
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here: Link [Answer is No here if your project does not utilize ML and AI]		X
Q5	Does this project involve external parties, funders, etc		X